# Dachuan Liu

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*Research interests: Multi-Agent Path Finding (MAPF), Multi-Robot Planning & Coordination, Self-Reconfigurable Modular and Swarm Robotics*

## Education

**University of Michigan** Ann Arbor, MI
*B.S.E. in Computer Engineering | GPA: 3.91/4.00* *Expected May 2027*
- Relevant coursework: Signals and Systems (EECS 216, A+), Data Structures and Algorithms (EECS 281), Embedded Control Systems (EECS 461), Computer Organization (EECS 370)

**Shanghai Jiao Tong University, UM-SJTU Joint Institute** Shanghai, China
*B.S.E. in Mechanical Engineering (Dual Degree) | GPA: 3.65/4.00* *Expected Aug 2027*
- Relevant coursework: Design and Manufacturing, Mechanical Laboratory, Differential Equations and Linear Algebra

## Publications

[C1] **D. Liu**, S. Zhou, S. Zhao, Z. Ren. "LSRP*: Scalable and Anytime Planning for Multi-Agent Path Finding with Asynchronous Actions." *Under review*, IEEE International Conference on Robotics and Automation (ICRA), 2027.

## Academic Service

- Sub-reviewer, *IEEE Transactions on Intelligent Transportation Systems* (T-ITS), 2026

## Research Experience

**Robotics, Autonomy, and Planning Lab, Shanghai Jiao Tong University** Shanghai, China
*Research Assistant | Advisor: Prof. Zhongqiang Ren* *Apr 2026 – Present*
- Developed **LSRP***, an anytime C++ solver for MAPF with Asynchronous Actions (MAPF-AA) that uses dominance pruning and restart strategies to keep improving solution quality within the time budget. Scales to 1,000 agents and reduces sum of costs by over 40% compared with LSRP; first-author paper submitted to ICRA 2027.
- Developed **LSRP*-LA**, extending LSRP* to large agents with $1\times1$ to $3\times3$ footprints, with a swept-area occupancy model and a rule layer that uses recursive push trees and lane evacuation to improve scalability and solution quality. Scales beyond 800 agents and reduces sum of costs by up to 62% compared with Layered LA-LaCAM.

**Self-Reconfigurable Modular Robotics, Robotics Department, University of Michigan** Ann Arbor, MI
*Undergraduate Researcher (ROB 490) | Advisor: Prof. Steven Ceron* *Sep 2026 – Present*
- Finalizing the design of, and fabricating, a fleet of 20–25 cylindrical modular robots that attach to neighbors with an extendable, rotating internal magnet and reconfigure by rolling around each other's perimeters; working with PhD student Tianyi Hu.
- Diagnosed a stall in the self-rotation drive (insufficient motor torque under shell friction) and specified a dual high-torque motor configuration.
- Redesigned the onboard power system after identifying that a 3.7 V cell could not meet the motor's 1.47 A stall current; selected a 7.4 V 2S LiPo with a 3 A DC-DC buck converter regulated to 5–6 V.
- Ran 200-cycle fatigue tests of the magnet extension/retraction gear mechanism (no observable wear) and fabricated a replacement lead screw; module assembly time is 2–3 hours each.

**Multidisciplinary Design Program, University of Michigan** Ann Arbor, MI
*Simulation Engineer | Advisor: Prof. Paul Green* *Jan 2026 – Present*
- Designed driving-simulation experiments in Unreal Engine 5 that vary obstacle type and curve radius, and analyze drivers' visual demand using the visual occlusion method.
- Running participant experiments and data analysis for a manuscript in preparation.

**Software Engineering Research Group, Shanghai Jiao Tong University** Shanghai, China
*Software Engineer | Advisor: Prof. Yan Liu* *Feb 2024 – Oct 2024*
- Built a full-stack interactive teaching platform (HTML/CSS/JS, MySQL) for teacher–student interaction and research data management; implemented the student-side client, which received a registered software copyright.

## Teaching Experience

**EECS 216: Introduction to Signals and Systems, University of Michigan** Ann Arbor, MI
*Instructional Aide* *Sep 2026 – Dec 2026*
- Lead 5 lab sections (30 students each), hold weekly office hours, and grade lab reports and exams.

**ECE 216: Introduction to Signals and Systems, UM-SJTU Joint Institute** Shanghai, China
*Teaching Assistant | Instructor: Prof. Heng Qiao* *May 2026 – Aug 2026*

- Taught twice-weekly recitations and weekly office hours; led 3 lab sessions (30 students) and graded assignments, lab reports, and exams.

**MATH 286: Differential Equations and Linear Algebra, UM-SJTU Joint Institute** Shanghai, China
*Teaching Assistant | Instructor: Prof. Horst Hohberger* *Sep 2025 – Dec 2025*

- Taught weekly recitations covering linear algebra, ODEs, Fourier series/transforms, and Laplace transforms; graded weekly assignments, four quizzes, and two exams.

## Selected Projects

**Learning Priority Orders for Prioritized SIPP in MAPF-AA (ECE 7606)** Shanghai, China
*C++, Reinforcement Learning* *May 2026 - Aug 2026*

- Trained a Plackett-Luce ranking policy with REINFORCE over per-agent conflict features to order agents for prioritized SIPP in MAPF with asynchronous actions; on held-out four maps (den312d, Room64, Warehouse, Random32) instances, raised success rate at 60 agents from 35% to 65% and the largest solved team from 80 to 140 agents.

**Drone Delivery Route Optimization (EECS 281)** University of Michigan
*C++* *Sep 2025 – Dec 2025*

- Solved TSP-style delivery routing on graphs: Prim's MST for connectivity, branch-and-bound with pruning for exact solutions, and arbitrary insertion + 2-opt for near-optimal solutions under strict runtime limits.

**Column-Climbing Robot (ME 350)** UM-SJTU Joint Institute
*SolidWorks, Raspberry Pi, Python* *May 2025 – Aug 2025*

- Designed and fabricated a robot that climbs columns of varying diameter; invented a bicycle-clamp locking mechanism that holds position without power, and built the motor and laser-sensor control loop.

## Skills

**Programming:** C++, Python, C, MATLAB, SQL, HTML/CSS/JavaScript
**Robotics & Tools:** Simulink, PyTorch, Unreal Engine 5, Git, LabVIEW, Mathematica, Flask
**Hardware:** SolidWorks, Arduino, Raspberry Pi, NI myDAQ
**English:** TOEFL iBT 107

## Honors and Awards

JI Student Development Scholarship, UM-SJTU Joint Institute 2023 – 2024
Outstanding Student Scholarship (Class B), Shanghai Jiao Tong University 2023 – 2024
Registered Software Copyright (teaching platform, student client) Sep 2024
Basic Teaching Assistant Certificate, UM-SJTU Joint Institute Aug 2024